Competition Problem: Create an n-type nanowire transistor. 2D simulation with cylindrical coordinates should be used. While this is a nanowire, we will use some nanosheet terminologies. For beginners, this is also an important learning process. You are advised to learn the new terms using LLMs or search engines. Optimize the transistors to achieve the best metrics given below, with the following constraints

1) The structure can be created through geometric operations or process simulation
2) Assume 1GPa stress in the Source/Drain direction. Turn on the appropriate stress models related to mobility and band structure.
3) Equivalent oxide thickness should 0.9nm (gate insulator stack design should be reasonable)
4) L_G (gate metal length) = 15nm; diameter = 6nm
5) Use metal gate with workfunction = 4.4eV
6) T = 300K
7) V_{DD} = 0.7V
8) Inner spacer = 4nm
9) Users have the freedom to engineer any source/drain and doping under the inner spacer (with realistic doping). Channel should be undoped (or unintentionally doped at $10^{15}cm^{-3}$).
10) Maximum concentration at any point should be less than $5×10^{20}cm^{-3}$.
11) Band-to-band tunneling needs to be turned on to account for leakage. No need to turn on gate tunneling.
12) Quantum corrections should be turned on.
13) Use drift-diffusion for transport. The maximum carrier velocity should be realistic for the model used. This is inaccurate, but we want to make the problem less difficult. In reality, if DD is used, calibrated ballistic correction should be used.
14) Appropriate mobility models should be turned on.

Metrics: Participants need to provide the following numbers in their submission.

a) I_{ON} (defined by $V_D = V_{DD}$ and $V_G = V_{DD}$ in A)
b) I_{OFF} (defined by $V_D = V_{DD}$ and $V_G = 0$ in A)
c) Peak g_m (at $V_D = V_{DD}$ in S)
d) Maximum electron velocity in the transistor at $V_D = V_G = V_{DD}$ in cm/s

Submission: Participants will be disqualified if any of the following is missing

1) Fill in the 4 metrics in the form provided.
2) Submit a **single** file (<10MB) that is ready to run. Sentaurus users must submit the whole project in swb gzip format (P-2019.03 release or newer should be used). Participants must make sure the project can be decompressed. The projects should be ready to run without the need for external scripts or environment settings. Silvaco users must submit a single input deck text file (so it creates the structure and loads into the electrical

simulation. No supplementary files are allowed. 2025 Baseline or later should be used.) Submissions that are not ready to run will be disqualified.

3) A PDF summary (< 3 pages, <10MB) to describe the design and the simulation results. Report clarity is evaluated as a part of the competition. The report should at least include the following:
    a. Snapshots of the structure showing the materials, meshing, and doping.
    b. Explanations of how the constraints are met.
    c. Snapshots of the $I_D V_G$ curves.
    d. Calculation of EOT should be detailed.

Submission Process:

Participants should submit the results before the deadline to https://ieeeforms.wufoo.com/forms/x14wp1fx19tzaep/ . Any submission after the deadline will be ignored. If the participants have a problem using the submission portal, please email the results to hiuyung.wong@sjsu.edu.